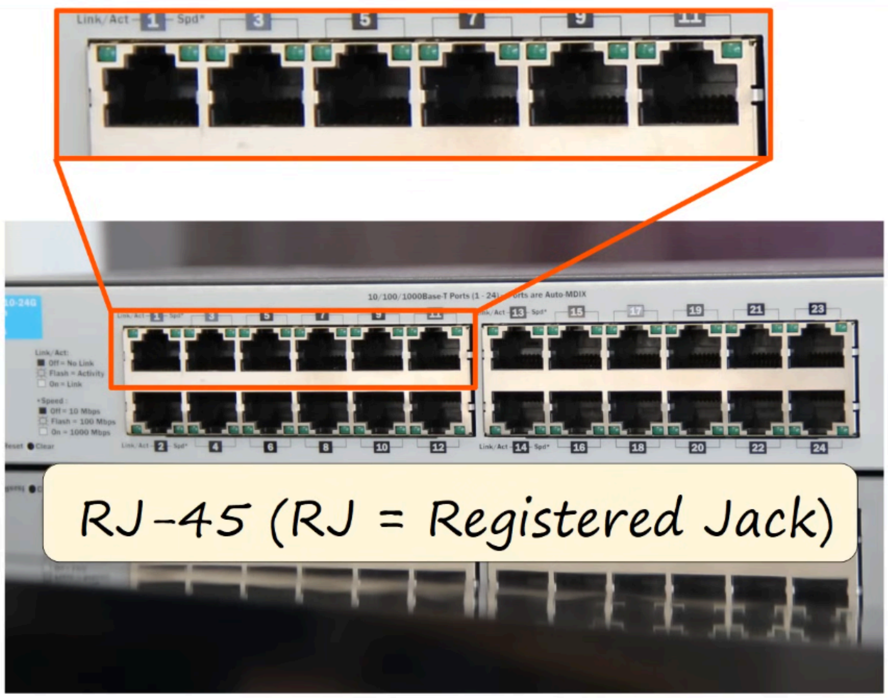
Day 2: Interfaces And Cables

SWITCHES provide many PORTS for connectivity (usually 24)

These PORTS tend to be RJ-45 (Registered Jack) ports.



WHAT IS ETHERNET?

- Ethernet is a collection of network protocols/standards.

Why do we need network protocols and standards?

- provide common communication standards over networks.
- provide common hardware standards to allow connectivity between devices.

Bits and Bytes:

- Connections between devices operates at a set speed.
- These speeds are measured in "bits per second" (bps, Kbps, Mbps,...), not “bytes per second”.
- A bit is a value of "0" or "1". A byte is 8 bits (0s and 1s)

Size	**# of Bits**
1 kilobit (Kb)	1,000

Size	# of Bits
1 megabit (Mb)	1,000,000
1 gigabit (Gb)	1,000,000,000
1 terabit (Tb)	1,000,000,000,000

Ethernet standards are:

- Defined in the IEEE 802.3 standard in 1983
- IEEE = Institute of Electrical and Electronics Engineers

ETHERNET STANDARDS (COPPER)

Speed	Common Name	IEEE Standard	Informal Name	Maximum Length
10 Mbps	Ethernet	802.3i	10BASE-T	100m
100 Mbps	Fast Ethernet	802.3u	100BASE-T	100m
1 Gbps	Gigabit Ethernet	802.3ab	1000BASE-T	100m
10 Gbps	10 Gigabit Ethernet	802.3an	10GBASE-T	100m

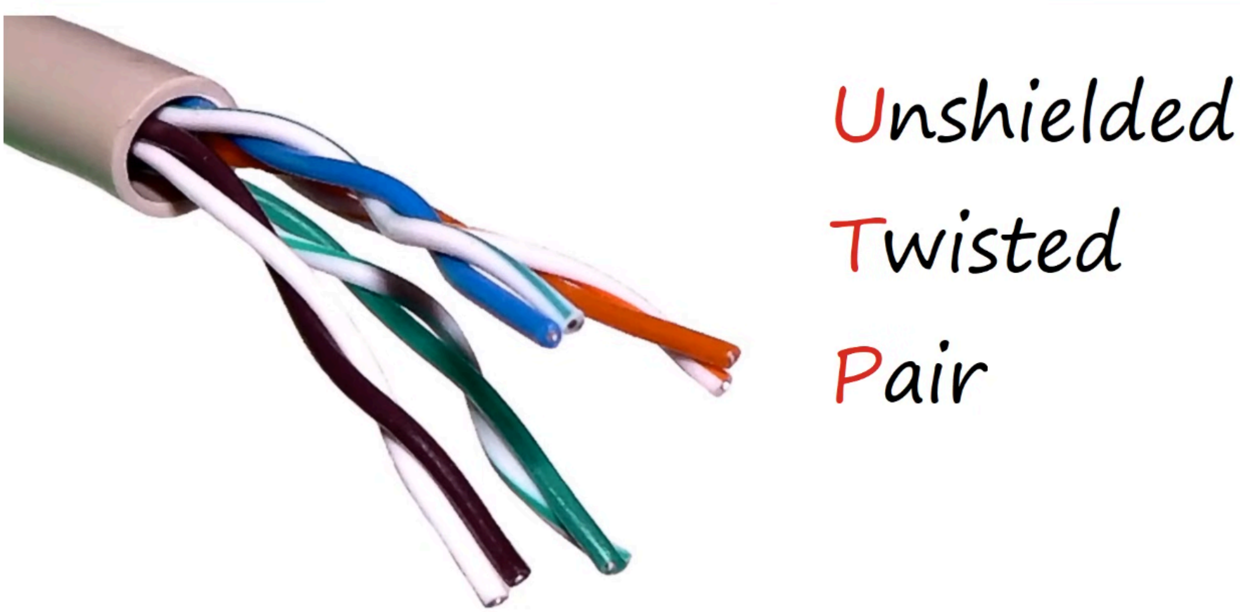
BASE = refers to Baseband Signaling

T = Twisted Pair

Most Ethernet uses copper cables.

UTP Cables

- UTP or Unshielded Twisted Pair (no metallic shield) Twist protects against EMI (Electromagnetic Interference)



1000BASE-T and 10GBASE-T = 4 pairs (8 wires)

10BASE-T and 100BASE-T = 2 pairs (4 wires)

HOW DO DEVICES COMMUNICATE VIA THEIR CONNECTIONS?

Each ethernet cable has a RJ-45 plug with 8 pins on the ends:

- PCs Transmit(TX) data on Pins #1-2
- Switches Receive(RX) data on Pins #1-2
- PCs Receive(RC) data on Pins #3,6
- Switches Transmit(TX) data on Pins #3,6

This allows **Full-Duplex** transmission of data.

Full-duplex transmission allows two devices to send and receive data simultaneously without waiting for each other. By utilizing two dedicated channels (or by dividing bandwidth/frequencies), it eliminates turn-based delays and collisions, doubling overall throughput.

What if a Router / Switch connect?

- Routers Transmit(TX) data on Pins #1-2
- Routers Receive(RX) data on Pins #3,6
- Switches Transmit(TX) data on Pins #3,6
- Switches Receive(RX) data on Pins #1-2

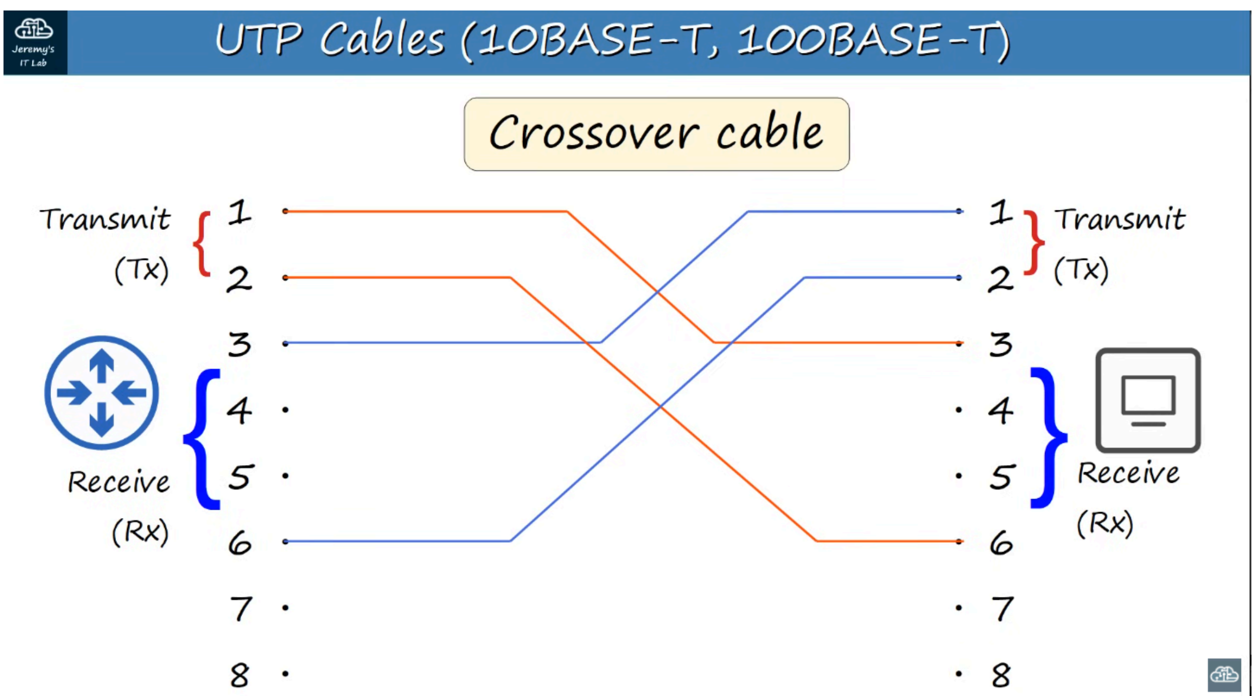
Routers and PCs connect the same way with Switches.

The cable used to connect is called a "Straight-Through" cable.

What if we want to connect similar devices to each other? For example, Router to Router - Pc to Pc - Switch to Switch - Pc to Router?

We CANNOT use a "Straight-Through" cable. We MUST use a "Crossover" cable.

This cable swaps the pins on one end to allow connection to work.

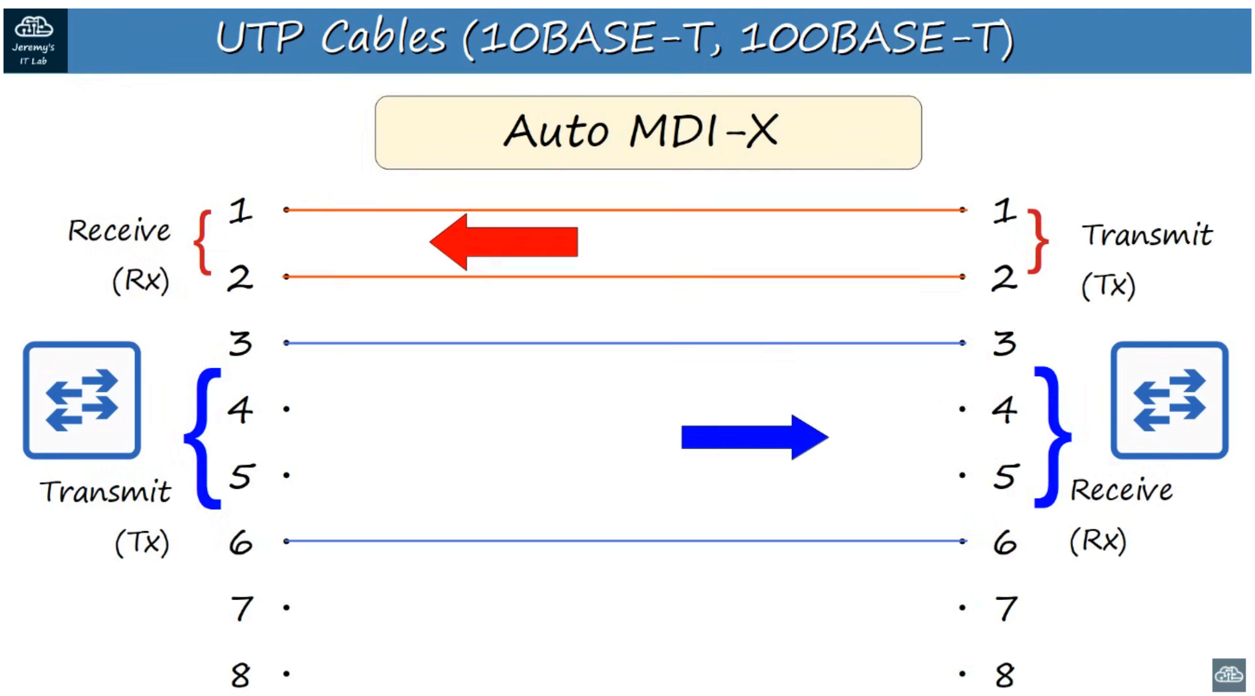


PIN#1 --⟶ PIN#3 PIN#2 --⟶ PIN#6

PIN#3 --⟶ PIN#1 PIN#6 --⟶ PIN#2

DEVICE TYPE	TRANSMIT (Tx) PINS	RECEIVE (Rx) PINS
ROUTER	1 and 2	3 and 6
FIREWALL	1 and 2	3 and 6
PC	1 and 2	3 and 6
SWITCH	3 and 6	1 and 2

Most modern equipment now has AUTO MDI-X which **automatically detects** which pins their neighbour is transmitting on and adjust the pins they receive data on.



1000BASE-T/10GBASE-T = 4 pairs (8 wires)

Each wire pair is **bidirectional** so can transmit/receive much faster than 10/100BASE-T.

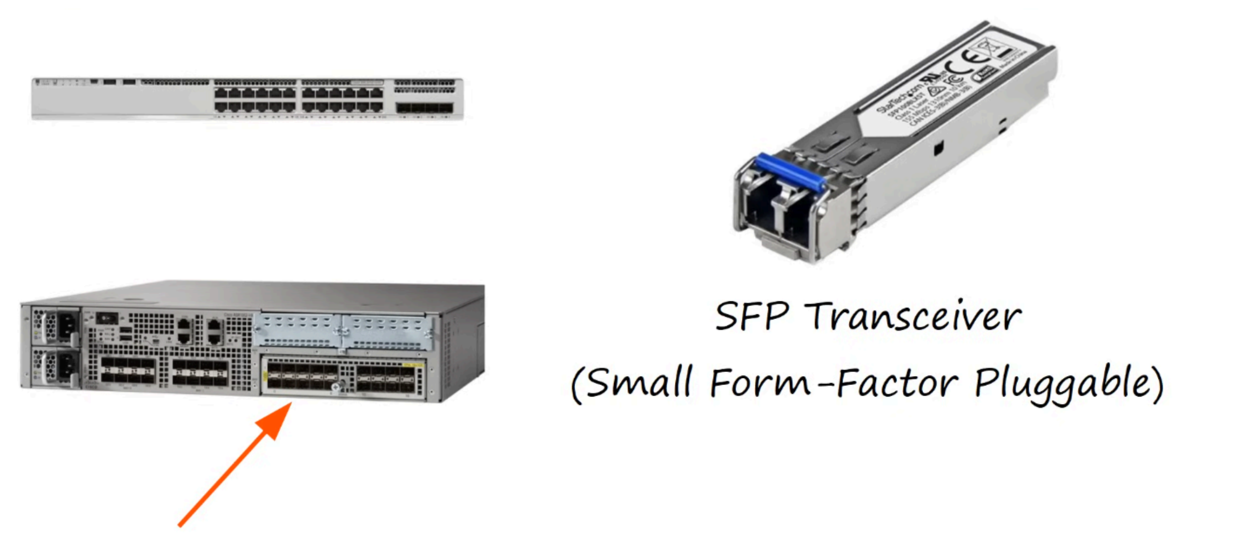
Bidirectional communication in cables is the ability to transmit data or signals in both directions simultaneously or in alternating sequence over the same physical wire or fiber.

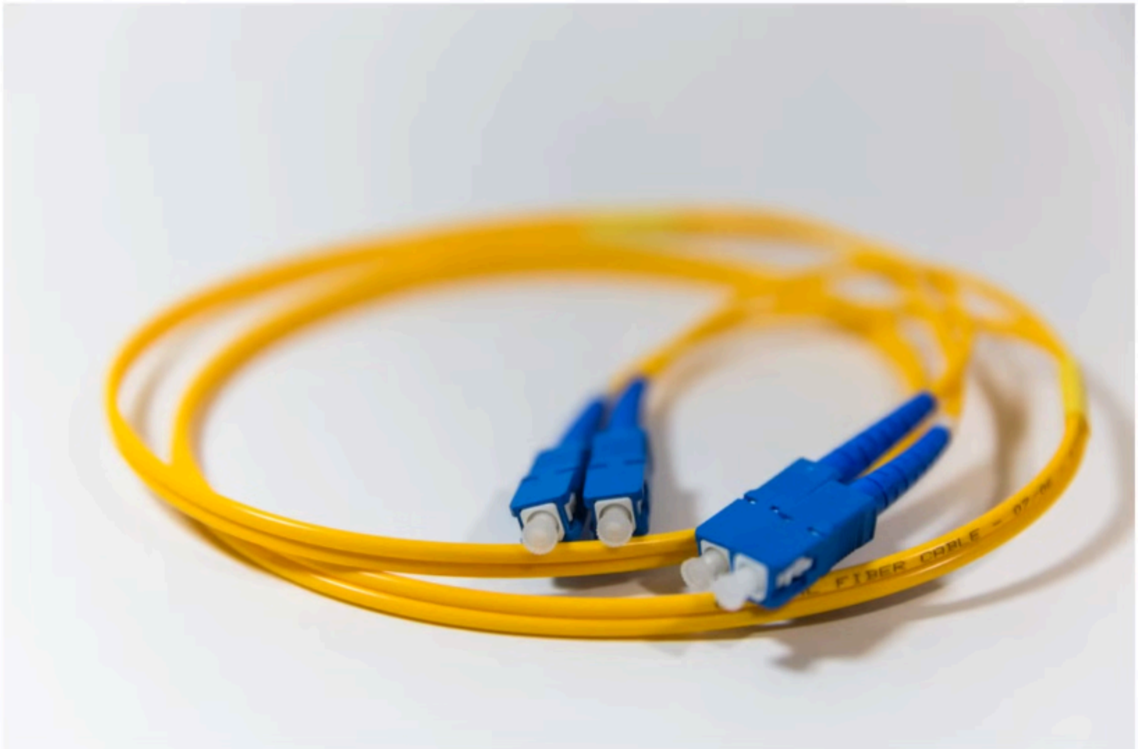
Fiber-Optic Connections:

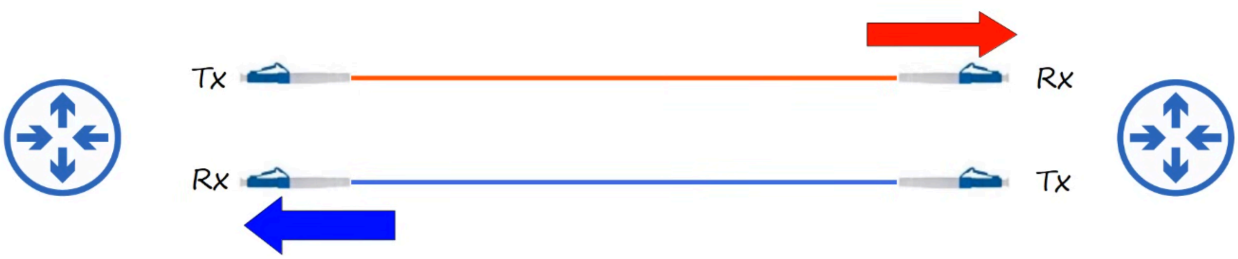
- Defined in the IEEE 802.3ae standard

SFP Transceiver (Small Form-Factor Pluggable) allows fiber-optic cables to connect to switches/routers.

- Have separate cables to transmit / receive.







4 parts to a fiber-optic cable.

There are TWO types of fiberoptic cable.

Single-Mode:

- Narrower than multimode
- Lighter enters at a single angle (mode) from a laser-based transmitter.
- Allows longer cables than both UTP and multimode fiber.
- More expensive than multimode fiber (due to more expensive laser-based SFP transmitters)

Multimode:

- Core is wider than Single-mode
- Allows multiple angles (modes) of light waves to enter core
- Allows longer cables than UTP but shorter than single-mode
- Cheaper than single-mode fiber (due to cheaper LED-based SFP transmitter)

Fiber Optic Standards:

Speed	**Standard**	**Connection Speed**	**Mode Support**	**Maximum Length**
1000BASE-LX	802.3z	1 Gbps	Multimode / Single	550 meters (Multi) / 5km (Single)
10GBASE-SR	802.3ae	10 Gbps	Multimode	400 meters
10GBASE-LR	802.3ae	10 Gbps	Single	10 kilometers
10GBASE-ER	802.3ae	10 Gbps	Single	30 kilometers

UTP vs Fiber-Optic Cabling:

UTP are:

- Lower cost than fiber-optic.
- Shorter maximum distance than fiber-optic (~100m).
- Can be vulnerable to EMI (Electromagnetic Interference).
- RJ45 ports used with UTP are cheaper than SFP ports.
- Emit (leak) a faint signal outside of cable, which can be copied (security risk).

Fiber-Optic:

- Higher cost than UTP.
- Longer maximum distance than UTP.
- No vulnerability to EMI.
- SFP ports are more expensive than RJ45 ports (single-mode is more expensive than multimode).
- Does not emit any signal outside of the cable (no security risk).